

Phân công:

- 2 topics (Association Rules Mining, Recommender System): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu) + Section 3: Thảo
- 3 topics (Fuzzy Logic, Linear Regression, Logistic Regression): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu): Nguyễn
- 3 topics (Latent Dirichlet Allocation, Deep Neural Networks, Word Embedding): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu): Lịch

POST-TEST

SECTION 1:

Association Rules Mining

No.	Bloom's Level	Question
1	Remembering	Which evaluation metric represents the proportion of the total transactions in the database that contain both itemset X and itemset Y simultaneously? A. Confidence B. Support C. Lift D. Conviction
2	Remembering	According to the downward-closure property of support (the Apriori Principle), if an itemset is found to be infrequent (below minSup), what can be strictly concluded about all of its supersets? A. All of its supersets must be frequent.

		B. All of its supersets must also be infrequent and can be safely pruned. C. The confidence of its supersets will automatically equal 100%. D. Its supersets can only be mined using a non-transactional database scan.
3	Understanding	Why does the Apriori algorithm perform a specific "Prune step" during candidate generation (Apriori-gen) by checking the frequency of all (k-1)-subsets of a candidate k-itemset? A. To verify that the candidate has a higher statistical variance than its subsets. B. Because if any (k-1)-subset of a candidate k-itemset is infrequent, the candidate itself cannot possibly be frequent, allowing it to be dropped without checking the database. C. To compress the size of the database by deleting transactional noise. D. To allow the algorithm to generate rules with low confidence levels.
4	Understanding	The Association Rules Mining problem is traditionally decomposed into two distinct sub-problems. What is the primary focus and computational bottleneck handled in the first sub-problem? A. Calculating the exact Lift and Conviction values for all possible rules. B. Discovering all frequent itemsets whose support count satisfies the minSup threshold. C. Generating rule implications from every arbitrary item permutation. D. Reducing the number of database items using dimensionality reduction.

5	Applying	Given a database of 500 grocery transactions. If "Milk" appears in 300 transactions, "Bread" appears in 200 transactions, and both "Milk" and "Bread" appear together in 150 transactions, calculate the exact Confidence of the association rule: {Milk} $\rightarrow$ {Bread}. A. 30% B. 50% C. 75% D. 40%
6	Applying	Utilizing the same database details from Question 5 (500 total transactions, 150 co-occurrences of Milk and Bread), what is the exact Support value for the itemset {Milk, Bread}? A. 30% B. 40% C. 50% D. 75%
7	Analyzing	What is the fundamental structural reason why the FP-Growth algorithm typically runs significantly faster than the Apriori algorithm on large-scale, highly dense transactional datasets? A. FP-Growth uses a simpler mathematical formula for calculating rule Lift. B. FP-Growth processes only categorical data and drops numerical attributes.

		C. FP-Growth compresses the transactional database into a frequent-pattern tree structure, eliminating candidate generation and requiring only two database scans. D. FP-Growth increases the minimum support threshold automatically during runtime.
8	Analyzing	Consider a frequent itemset X and its subset Y ($Y \subset X$). Compare the Support value of X and the Support value of Y. Which statement is mathematically always true? A. $\text{Support}(X) \geq \text{Support}(Y)$ B. $\text{Support}(X) \leq \text{Support}(Y)$ C. $\text{Support}(X) = \text{Support}(Y)$ D. Their support relationship depends entirely on the rule confidence.
9	Evaluating	A data analyst discovers an association rule that possesses an exceptionally high Confidence level of 95% but a Lift value of 0.82. How should the team critically evaluate the operational value of this rule? A. It is a highly reliable rule that must be deployed immediately because confidence is near perfect. B. The data is mathematically corrupted because Lift can never drop below 1.0 when confidence is high. C. The rule has little to no practical value because a Lift less than 1.0 indicates a negative correlation, meaning the presence of the antecedent actually reduces the likelihood of the consequent occurring. D. The team should decrease the minimum support threshold to make the Lift increase.

10	Creating	Suppose you are adapting the traditional association rules mining workflow for a medical safety system where discovering "Closed Itemsets" (itemsets where no immediate superset has the same support count) is required to reduce rule redundancy. How would you modify the candidate evaluation procedure? A. By dropping any candidate itemset that has a support count equal to any of its subsets. B. By adding a checking constraint to ensure that an itemset is only retained if all of its supersets possess a strictly smaller support count. C. By forcing the algorithm to skip the database scan step completely. D. By only generating rules that contain exactly a single item in the consequent.
----	----------	--

Recommender System

No.	Bloom's Level	Question
1	Remembering	Which accuracy metric calculates the square root of the average of squared differences between predicted ratings and actual user ratings, thereby placing a heavier penalty on larger prediction errors? A. Mean Absolute Error (MAE) B. Root Mean Square Error (RMSE) C. Precision at k (P@k) D. Mean Average Precision (MAP)

2	Remembering	In Content-based Filtering systems, what is the primary mathematical purpose of integrating the Inverse Document Frequency (IDF) component into TF-IDF term weighting? A. To normalize the length of item description vectors. B. To reduce the relative weight of generic words that appear across almost all items, highlighting unique descriptive terms. C. To calculate the baseline user rating bias across the system. D. To map continuous ratings into a discrete binary classification range.
3	Understanding	Why is Item-based Collaborative Filtering often preferred over User-based Collaborative Filtering in massive commercial web systems (like Amazon or Netflix)? A. User similarity profiles are inherently more static and scale better over time. B. Item relationships and similarity scores are more stable and the total number of items is often significantly smaller or changes slower than the shifting number of active users. C. Item-based systems completely bypass the cold-start problem for new users. D. Item-based approaches do not require any explicit rating input to make predictions.
4	Understanding	From a strategic product perspective, what is the core benefit of designing a recommender engine that successfully suggests niche, obscure items located in the "Long Tail" of the distribution? A. It ensures the system achieves a near-zero RMSE score. B. It optimizes system memory by filtering out highly active users. C. It uncovers hidden, personalized items that users would not easily discover on their own, enhancing user serendipity and long-term platform loyalty.

		D. It forces the system to operate strictly as a popularity-based model.
5	Applying	Suppose a user profile vector and an item feature vector are represented in a binary space. If the dot product of these two vectors yields a high value, what does this mathematically imply in a Content-based Filtering framework? A. The user has a high similarity score with other users who purchased this item. B. The item has attributes that heavily align with the specific features of items the user has liked or interacted with in the past. C. The item belongs to the most popular category across the entire platform. D. The rating matrix suffers from extreme data sparsity issues.
6	Applying	In a Memory-based Collaborative Filtering setup, User A and User B have a high Pearson Correlation Coefficient of 0.88. If User A highly rates a newly released item with 5 stars, what rating is the system most likely to predict for User B regarding this identical item? A. A very low rating (1 star) due to correlation inversion. B. A high predicted rating, adjusted against User B's baseline average rating behavior. C. No prediction can be computed because they must share content tags. D. A completely neutral rating (3 stars) to prevent overfitting.
7	Analyzing	What is the fundamental scalable disadvantage of using a pure "Memory-based" (Neighborhood) Collaborative Filtering approach in a production system containing millions of active users and items? A. It requires complex domain knowledge engineering to tag item features.

		B. It cannot process explicit numeric feedback like 1-5 star ratings. C. The entire rating matrix must reside in memory to compute similarities at runtime, making it computationally expensive and non-scalable as data grows. D. It forces the model to suffer from gradient vanishing during backpropagation.
8	Analyzing	How does the "Recursive Collaborative Filtering" approach attempt to mitigate the intense challenge of data sparsity within a highly sparse rating matrix? A. By switching the entire platform backend to a static popularity-based strategy. B. By iteratively predicting missing ratings for standard neighbors first, and then using those synthetic predictions as pseudo-ground truth inputs to calculate recommendations for the active user. C. By forcing users to rate at least 50% of the available item catalog. D. By dropping all items that belong to the tail end of the distribution curve.
9	Evaluating	An AI researcher evaluates a new music recommendation system and notes that while the raw offline validation accuracy (RMSE) is poor, live A/B testing yields extremely high user satisfaction and click-through rates. How can you explain this outcome? A. Accuracy metrics like RMSE are entirely uninformative in all machine learning tasks. B. The system may excel at explanation transparency, diversity, and delivering surprising yet relevant suggestions (Serendipity) which users value, despite missing the exact numerical rating targets. C. The live users are likely misinterpreting their own listening preferences. D. The validation dataset was completely empty due to a matrix factorization error.

10	Creating	Imagine you are designing a Hybrid Recommender System architecture specifically tailored to eliminate the "New User Cold Start" problem while maintaining high scalability. Propose the most robust combination of methodologies. A. Deploy a pure Matrix Factorization model that updates through batch jobs once a month. B. Initialize a Knowledge-based or Content-based questionnaire during user onboarding to capture explicit preferences, then seamlessly transition to a Model-based Collaborative Filtering system (e.g., SVD) once the user accumulates interaction data. C. Implement a pure User-User Collaborative Filtering model that uses Cosine similarity on an uncentered matrix. D. Use an unregularized Deep Neural Network that accepts only one-hot encoded user IDs.
----	----------	--

Fuzzy Logic

No.	Bloom's Level	Question
1	Remembering ▾	What is the term for a mathematical function used in Fuzzy Logic to define the degree of truth of a statement, mapping input values to the interval [0, 1]? A. Crisp Function B. Transfer Function C. Membership Function D. Probability Density Function
2	Remembering ▾	The Mamdani-type Fuzzy Inference System typically uses which operator for the implication step (applying the rule strength to the output fuzzy set)? A. Min (or product) B. Max C. Sum D. Average

3	Understanding ▾	Why is the Center of Gravity (Centroid) method generally considered the most robust defuzzification technique? A. It always selects the mean of the maximum output value. B. It considers the entire shape and area of the aggregated fuzzy output set. C. It only requires checking the peak value of the output fuzzy set. D. It avoids the need for a rule base entirely.
4	Understanding ▾	Explain the consequence of using the Bounded Sum ($A \oplus B = \min(1, \mu_A(x) + \mu_B(x))$) as the S-norm (OR operator) compared to the standard Max operator. A. Bounded Sum always results in a lower membership degree. B. Bounded Sum can allow the resulting membership degree to exceed $\max(\mu_A, \mu_B)$, potentially reaching 1 faster when the two sets overlap significantly. C. Bounded Sum is only used in Sugeno-type systems. D. Bounded Sum is equivalent to the Max operator in all cases.
5	Applying ▾	A system has two rules contributing to the output 'Strong Wind': Rule 1 firing strength is 0.7, and Rule 2 firing strength is 0.9. If the output sets are aggregated using the standard Max (Union) operator, what is the height of the combined fuzzy set 'Strong Wind'? A. 0.7 B. 0.9 C. 1.6 D. 0.63
6	Applying ▾	Given a fuzzy set $A = \{ (1, 0.2), (2, 0.8), (3, 0.5) \}$ and a fuzzy set $B = \{ (1, 0.6), (2, 0.3), (3, 0.9) \}$. Calculate the membership values for the intersection $A \cap B$ using the standard Min operator. A. $\{(1, 0.6), (2, 0.8), (3, 0.9)\}$ B. $\{(1, 0.8), (2, 1.1), (3, 1.4)\}$ C. $\{(1, 0.2), (2, 0.3), (3, 0.5)\}$ D. $\{(1, 0.4), (2, 0.5), (3, 0.4)\}$
7	Analyzing ▾	In a complex control system, if the number of input variables and the number of linguistic terms per variable both increase, what is the primary consequence concerning the rule base? A. The defuzzification process becomes simpler. B. The system becomes less sensitive to input changes. C. The rule base suffers from combinatorial explosion (exponential growth in the number of rules). D. The Fuzzification step requires less computation.

8	Analyzing ▾	Compare a Mamdani FIS that uses the Min operator for implication and the Center of Gravity (COG) for defuzzification versus a Sugeno FIS using weighted average. Which system is generally faster for real-time control, and why? A. Mamdani, because its output calculation is simpler. B. Mamdani, because it uses fuzzy sets directly. C. Sugeno, because it avoids the computationally intensive integration required by the COG defuzzification method. D. Sugeno, because it produces fuzzy outputs.
9	Evaluating ▾	Evaluate the critical trade-off when selecting a narrow (steep slope) versus a wide (gentle slope) Membership Function for the linguistic term "High" in a temperature controller. A. A narrow MF increases the robust nature of the system. B. A narrow MF increases the system's sensitivity to small input changes, potentially leading to oscillatory control. C. A wide MF decreases the overlap between linguistic terms, simplifying the rule base. D. The shape of the MF only affects the aesthetic appearance of the control system.
10	Creating ▾	Design a fuzzy rule that addresses the concept of hedging or uncertainty by linking a highly certain input state with a moderately conservative output action. Use the linguistic variables: Input_Certainty (Very_High, Moderate) and Output_Action (Aggressive, Cautious, Neutral). A. IF Input_Certainty IS Very_High THEN Output_Action IS Aggressive. B. IF Input_Certainty IS Moderate THEN Output_Action IS Neutral. C. IF Input_Certainty IS Very_High THEN Output_Action IS Cautious. D. IF Input_Certainty IS Low THEN Output_Action IS Aggressive.

Linear Regression

Linear Regression

No.	Bloom's Level	Question
1	Remembering ▾	What is the primary goal of Simple Linear Regression? A. To find the best fitting curve. B. To model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line. C. To classify data points into discrete categories. D. To minimize the sum of absolute errors.

2	Remembering ▾	In the linear regression equation $Y = \beta_0 + \beta_1 X + \epsilon$ represent? A. The Y-intercept. B. The slope of the regression line. C. The residual error. D. The expected value of Y when X is zero.
3	Understanding ▾	What is the fundamental difference between Simple Linear Regression and Multiple Linear Regression? A. Simple Linear Regression has continuous variables; Multiple Linear Regression has categorical variables. B. Simple Linear Regression uses one predictor variable; Multiple Linear Regression uses two or more predictor variables. C. Multiple Linear Regression is only used for non-linear relationships. D. Simple Linear Regression cannot handle noise (ϵ).
4	Understanding ▾	Explain the meaning of a coefficient of determination (R^2) value of 0.85 in a model. A. 85% of the data points fall directly on the regression line. B. 85% of the variance in the dependent variable (Y) is predictable from the independent variable (X). C. The model has an 85% chance of predicting the correct value. D. The correlation between X and Y is 0.85.
5	Applying ▾	A model predicts house price (Y) based on square footage (X) with the equation $\text{Price} = 50,000 + 100 \cdot \text{SqFt}$. If a house is 2,000 SqFt, what is the predicted price? A. \$250,000 B. \$200,000 C. \$150,000 D. \$100,000

6	Applying ▾	Which loss function is most commonly minimized during the training of a standard Linear Regression model using the Ordinary Least Squares (OLS) method? A. Mean Absolute Error (MAE) B. Binary Cross-Entropy C. Mean Squared Error (MSE) D. Log Loss
7	Analyzing ▾	When performing residual analysis, if you observe a distinct pattern (e.g., a parabolic shape) in the plot of residuals versus fitted values, what does this indicate about the linear regression model? A. The model is highly accurate. B. The assumption of linearity has been violated. C. The data is perfectly correlated. D. The residuals are normally distributed.
8	Analyzing ▾	Why is it crucial to check for multicollinearity in Multiple Linear Regression, and what statistical measure is often used to detect it? A. Multicollinearity makes the model faster; VIF is used. B. Multicollinearity inflates the standard errors of the coefficients, making them unstable; Variance Inflation Factor (VIF) is used. C. Multicollinearity is only a problem in simple regression; R-squared is used. D. Multicollinearity implies a perfect fit; ANOVA is used.

9	Evaluating ▾	Evaluate the trade-off between using a high-degree polynomial regression model (which perfectly fits the training data) versus a simple linear regression model for future prediction. Which model is generally preferred for generalization, and why? A. The simple linear regression model, because the polynomial model is prone to overfitting and will perform poorly on new, unseen data. B. The polynomial model, because a perfect fit on the training data guarantees high accuracy everywhere. C. The choice depends only on the R-squared value, not on generalization. D. Both models generalize poorly; only non-parametric models should be used.
10	Creating ▾	Propose a hypothetical feature transformation (e.g., squaring, logging) for the independent variable X in a dataset where the relationship between X and Y is initially exponential, and briefly explain why this transformation is necessary for Linear Regression. A. Transform X to log(X). This is necessary because Linear Regression assumes a linear relationship, and the logarithmic transformation effectively linearizes the initial exponential relationship. B. Transform X to X². This is necessary to satisfy the normality assumption of the errors. C. Transform Y to Y-1. This is necessary to deal with heteroscedasticity. D. No transformation is necessary; Linear Regression can handle exponential relationships directly.

Logistic Regression

Logistic Regression

No.	Bloom's Level	Question
-----	---------------	----------

1	Remembering ▾	What is the primary difference between Linear Regression and Logistic Regression regarding the dependent variable (Y)? A. Linear Regression predicts continuous values; Logistic Regression predicts probabilities for binary outcomes. B. Linear Regression uses categorical inputs; Logistic Regression uses continuous inputs. C. They both predict continuous values but use different optimization methods. D. Logistic Regression cannot handle multiple inputs.
2	Remembering ▾	Which function is applied in Logistic Regression to map the output of the linear equation into a probability range between 0 and 1? A. Rectified Linear Unit (ReLU) B. Softmax Function C. Sigmoid Function D. Tanh Function

3	Understanding ▾	Why does Logistic Regression use the Sigmoid function instead of directly using the linear equation $Y = \beta_0 + \beta_1 X + \epsilon$? A. To ensure the predicted outcome is non-negative. B. To transform the output into a value that can be interpreted as a valid probability (between 0 and 1). C. To minimize the Mean Squared Error (MSE). D. To correct for multicollinearity.
4	Understanding ▾	In Logistic Regression, the log-odds (logit) is defined as the logarithm of the ratio of the probability of success (p) to the probability of failure ($1 - p$). What is the formula for the log-odds? A. $\ln(p/(1-p))$ B. $p + (1-p)$ C. $1/p$ D. e^p

5	Applying ▾	A Logistic Regression model provides a log-odds (logit) value of 0 for a specific customer. What is the calculated probability (p) that this customer belongs to the positive class? A. 0 B. 0.5 C. 1 D. 0.731
6	Applying ▾	Which mathematical function is typically minimized as the cost function (loss function) when training a Logistic Regression model? A. Mean Squared Error (MSE) B. R^2 C. Cross-Entropy Loss (or Log Loss) D. Sum of Absolute Errors (SAE)

7	Analyzing ▾	How should a coefficient β be interpreted in Logistic Regression? A. A one-unit increase in X increases the predicted probability p by β. B. A one-unit increase in X increases the predicted Y by β. C. A one-unit increase in X increases the log-odds of the positive class by β. D. A one-unit increase in X decreases the residual error by β.
8	Analyzing ▾	In a classification dataset, if the data points are perfectly separated by a linear boundary, why might training a standard (unregularized) Logistic Regression model lead to numerical issues? A. The model will converge immediately. B. The coefficients (β) will approach infinity, indicating complete separation (perfect fit). C. The Cross-Entropy Loss will become zero, halting the training process prematurely. D. The Sigmoid function will output only 0s.

9	Evaluating ▾	For a system designed to predict whether a manufacturing machine will fail (positive class) or run normally (negative class), which metric should be prioritized and why? A. Precision, because minimizing False Positives is crucial for cost savings. B. Recall, because minimizing False Negatives (missed failures) is essential to prevent costly operational damage. C. Accuracy, because the overall correctness matters most. D. F1 Score, because it balances Precision and Recall equally.
10	Creating ▾	Propose a specific architecture or method to adapt a Binary Logistic Regression model for a problem with three or more mutually exclusive classes (e.g., predicting Low, Medium, or High risk levels). A. Use the Kernel Trick to map data to a higher dimension. B. Use Simple Linear Regression for each class independently. C. Use Multinomial Logistic Regression (Softmax Regression) or the One-vs-Rest (OvR) approach. D. Use the Sigmoid function multiple times on the same output.

Latent Dirichlet Allocation

No.	Bloom's Level	Question
-----	---------------	----------

1	Remembering ▾	In LDA, what do the hyperparameters α (alpha) and β (beta) represent? A. Number of topics and vocabulary size B. Smoothing parameters for topic-document and word-topic distributions C. Learning rate and momentum D. Number of iterations
2	Remembering ▾	Which sampling method is most commonly used for inference in LDA? A. MCMC with Gibbs Sampling B. Gradient Descent C. Backpropagation D. Expectation Maximization only
3	Understanding ▾	Why is LDA considered a generative probabilistic model? A. It predicts document labels directly B. It models how documents can be generated from topics and words C. It classifies documents into fixed categories D. It minimizes reconstruction error
4	Understanding ▾	What is the main difference between LDA and pLSA? A. LDA uses Dirichlet priors while pLSA does not B. pLSA is Bayesian while LDA is not C. LDA only works with long documents D. pLSA supports multiple topics per document
5	Applying ▾	If a document has topic distribution $\theta = [0.6, 0.3, 0.1]$, what does this mean? A. The document belongs to only one topic B. 60% of the words are likely generated from the first topic C. The document has exactly 3 topics D. The document is labeled with the dominant topic
6	Applying ▾	What happens when the number of topics (K) is set too high in LDA? A. Topics become more coherent B. Topics become fragmented and less interpretable C. Model training becomes faster D. Perplexity always decreases
7	Analyzing ▾	Which of the following is a major limitation of LDA? A. It cannot handle polysemy well B. It requires labeled data C. It captures word order effectively D. It works only with image data

8	Analyzing ▾	What is a key ethical concern when using LDA on private user messages? A. High computational cost B. Privacy violation and lack of consent C. Low model accuracy D. Difficulty in choosing K
9	Evaluating ▾	When is LDA most suitable for a task? A. Real-time sentiment analysis B. Topic discovery in large unlabeled text collections C. Image classification D. Time-series prediction
10	Creating ▾	Which technique is most effective to improve privacy in an LDA-based user profiling system? A. Increase number of topics B. Apply Differential Privacy or document anonymization C. Use larger vocabulary D. Reduce the number of iterations

Deep Neural Networks

No.	Bloom's Level	Question
1	Remembering ▾	What is the main benefit of using Residual Connections in deep networks? A. Reduces model size B. Increases activation values C. Helps mitigate vanishing gradient problem D. Faster inference speed
2	Remembering ▾	Which activation function is most commonly used in modern deep neural networks? A. Sigmoid B. Linear C. Softmax D. ReLU
3	Understanding ▾	Why is Batch Normalization commonly used? A. To replace activation functions B. To stabilize and speed up training C. To increase model capacity

		D. To reduce the number of layers
4	Understanding ▾	What does the vanishing gradient problem cause? A. Loss becomes infinite B. Weights update too quickly C. Gradients become very small in early layers D. Model converges too fast
5	Applying ▾	What should you do first if you observe exploding gradients? A. Add more layers B. Increase learning rate C. Remove Batch Normalization D. Apply Gradient Clipping
6	Applying ▾	If your DNN is severely overfitting, which technique is most effective? A. Add more neurons B. Increase learning rate C. Reduce batch size to 1 D. Use Dropout and Weight Decay
7	Analyzing ▾	Why have Transformers largely replaced RNNs in many tasks? A. Transformers use less data B. RNNs require more memory C. Transformers are easier to implement D. Transformers handle long-range dependencies better via self-attention
8	Analyzing ▾	What is the main risk when a DNN is used for hiring decisions? A. Slow inference time B. High computational cost C. Algorithmic bias from training data D. Overfitting on small data
9	Evaluating ▾	What is the biggest concern with highly accurate but black-box DNNs in healthcare? A. Training time B. Number of parameters C. Model size D. Lack of interpretability and trust
10	Creating ▾	Which approach is best to reduce gender bias in a DNN for loan approval? A. Use deeper architecture B. Increase learning rate C. Use only one-hot encoding D. Apply fairness constraints and remove sensitive features

Word Embedding

No.	Bloom's Level	Question
1	Remembering ▾	What is the key advantage of FastText over Word2Vec? A. It only works with English B. It requires labeled data C. It uses subword information (n-grams) D. It is faster to train
2	Remembering ▾	Why do static embeddings (e.g. Word2Vec) struggle with polysemy? A. They require too much memory B. They are too large C. Each word has only one fixed vector regardless of context D. They cannot handle rare words
3	Understanding ▾	What is the main advantage of contextual embeddings (e.g. BERT) over static embeddings? A. They only capture syntax B. They do not need training C. They are smaller in size D. Word representations change based on sentence context
4	Understanding ▾	In the analogy “king - man + woman $\approx$?”, what is the expected result? A. Prince B. Girl C. Mother D. Queen
5	Applying ▾	How can you best evaluate the quality of word embeddings? A. By checking training loss only B. By measuring model size C. By counting vocabulary D. Using word analogy and similarity tasks
6	Applying ▾	What type of bias is shown if “doctor” is closer to “man” and “nurse” to “woman”? A. Technical bias from algorithm B. Evaluation bias C. Random noise D. Societal/gender stereotype bias from training data

7	Analyzing ▾	What is a major limitation of GloVe compared to BERT? A. GloVe produces static embeddings B. GloVe cannot handle English C. GloVe needs labeled data D. GloVe is slower to train
8	Analyzing ▾	Why is bias in word embeddings dangerous in real applications? A. It increases vocabulary size B. It makes training slower C. It can lead to unfair or discriminatory decisions in downstream tasks D. It reduces model accuracy
9	Evaluating ▾	Which method is most effective for reducing bias in word embeddings? A. Remove stopwords B. Increase vector dimension C. Use larger batch size D. Use debiasing techniques or balanced training data
10	Creating ▾	Which approach is best for creating good word embeddings for Vietnamese? A. Use only one-hot encoding B. Use standard English Word2Vec directly C. Reduce vocabulary size D. Train on large Vietnamese corpus with subword handling and diacritic support

SECTION 2:

CASE STUDY: SMART URBAN LOGISTICS AND GREEN DELIVERY FLEET (PROJECT ECO-ROUTING)

1. Detailed Background:

Project Eco-Routing is a next-generation intelligent logistics management system deployed in a metropolitan area characterized by high population density and complex traffic patterns. The strategic objective of the project is to globally optimize delivery routes, minimize carbon emissions from transport fleets, and elevate customer service satisfaction through real-time data analysis.

To address the intricate nature of urban logistics, **Project Eco-Routing** avoids standalone operations and instead implements a Hybrid Multi-stage Delivery Ecosystem (Man-Machine Collaboration):

- **Trunk-Line Stage:** The system commands a fleet of Autonomous Electric Vehicles (AEVs) responsible for consolidating large-scale parcels from the Central Hub and autonomously navigating major avenues to transport them to distributed "Satellite Depots" within residential areas. Equipped with state-of-the-art sensors, these AEVs handle main-road traffic scenarios efficiently but are physically incapable of entering narrow alleys or climbing apartment complexes.
- **Last-Mile Delivery Stage:** At the satellite depots, a network of Gig-Economy Shippers (human couriers) utilizing mobile applications takes over the parcels unloaded by the AEVs. They use motorbikes to execute the final delivery leg, maneuvering through the city's characteristic narrow alleys and high-rise buildings to hand over packages directly to customers and handle face-to-face interactions.

The central AI system acts as the "brain," continuously absorbing data from both entities to optimize order batching, dynamic routing, time-of-arrival estimation, and cross-selling service recommendations.

2. Detailed Primary Data Sources Eco-Routing's central AI infrastructure continuously ingests and processes four multimodal data streams, representing classic data structures in Computer Science:

1. **Interaction Matrix & Order Logs (Tabular/Transactional Data):** Stores historical customer purchase records; itemsets frequently co-purchased in a single invoice; coordinates of satellite depots; and specifically, a matrix of customer satisfaction scores (Scale 1–5 stars) rating the human couriers post-delivery, along with click/skip logs of recommended food/product attachments on the application interface.
2. **IoT Sensor Grid & Traffic Status (Time-Series & Qualitative Data):** Real-time telemetry (emitted every few seconds) from the AEV fleet, including remaining battery capacity, electric motor temperature, vehicle stream velocity, ambient humidity, and junction traffic density via radar sensors. This stream is integrated in parallel with **qualitative status reports** actively submitted by couriers via their driver app during transit (e.g., selecting labels such as "heavily congested," "moderately crowded," or "clear" at specific intersections).
3. **Customer Feedback & Call Center Logs (Unstructured Text Data):** A collection of tens of thousands of text transcripts from customer service calls (handling complaints about delayed deliveries, damaged items, or courier behavior); user review comments written on the app; and periodically updated market research reports analyzing urban logistics trends.
4. **Surveillance Feed (Vision Stream - Image/Video Data):** Continuous video feeds transmitted from dashcams integrated into the AEVs during transit or at loading docks. This data feeds computer vision tasks, such as identifying and bounding static/dynamic obstacles, recognizing traffic signs, detecting road surface anomalies (potholes, deep water puddles), and classifying pedestrian density to ensure absolute operational safety.

3. Learner's Task: Based on the operational context and multi-source data components of **Project Eco-Routing**, your objective is to apply theoretical knowledge and practical frameworks from your **02 selected topics** to solve the knowledge transfer scenarios presented in the questions below.

Association Rules Mining

Question 1 (Applying): An engineer runs the Apriori algorithm on the "Interaction Matrix & Order Logs" database to find items frequently co-purchased by citizens in a single order. Suppose the minimum support threshold is set to 2% and minimum confidence is set to 60%. The system discovers the following rule: {Fast Food} -> {Soft Drink} [Support = 3%, Confidence = 75%]. Which of the following statements is the most accurate interpretation of this result?

- A. Soft drinks are present in 75% of all delivery orders processed by the system.
- B. Fast food and soft drinks are purchased together in exactly 3% of all processed orders, and out of those who ordered fast food, 75% also ordered a soft drink.**
- C. The system should automatically increase the support threshold because 3% is too low to be statistically valid for any logistical decision.
- D. A courier has a 75% probability of being assigned a delivery route that contains both fast food and soft drinks.

Question 2 (Analyzing):

During peak rainy hours, traffic density surges and delivery delays occur frequently. The data team wants to mine association rules where the consequent (Y) is strictly fixed as the item {Delivery Delay} to find environmental factors leading to this state. If they significantly decrease the minimum support threshold (minSup) to capture rare extreme weather events, how will this modification affect the computational complexity of the Apriori algorithm?

- A. Computational complexity will decrease because the algorithm will skip the database scan step.
- B. Computational complexity will remain unchanged since thresholds only filter the final rules, not the candidate generation phase.
- C. Computational complexity will increase drastically because more itemsets will satisfy the lower threshold, leading to an explosion of candidate generation (Apriori-gen) and repetitive database scans.**
- D. The algorithm will immediately converge in the first iteration because rare events do not produce subsets.

Question 3 (Analyzing):

The operational team compares two association rules generated from the order history logs:

- Rule 1: {Fried Chicken} -> {French Fries} [Support = 10%, Confidence = 70%, Lift = 1.2]
- Rule 2: {Burger} -> {Napkins} [Support = 5%, Confidence = 85%, Lift = 0.95]

How should the team evaluate these two rules for a cross-selling campaign on the app?

- A. Rule 2 should be prioritized over Rule 1 because it has a higher confidence level (85% > 70%).
- B. Rule 2 is highly effective because a Lift close to 1 indicates a powerful, direct causal relationship between Burgers and Napkins.
- C. Rule 1 should be prioritized because Lift > 1 indicates a positive correlation, meaning ordering Fried Chicken actually increases the likelihood of ordering French Fries; whereas Rule 2 has a Lift < 1, implying Burgers and Napkins are negatively correlated or independent despite high confidence.**

D. Both rules are invalid because Support must always be strictly greater than Confidence in market basket analysis.

Question 4 (Evaluating):

To optimize space in a regional satellite depot, engineers want to discover "Closed Itemsets" from the parcel transaction logs (itemsets where no immediate superset has the exact same support count). How should they mathematically evaluate a candidate itemset C to confirm it qualifies as a Closed Itemset?

- A. By checking if the confidence of all rules generated from C is exactly 100%.
- B. By verifying that for every superset S (where S contains C), the support count of S is strictly less than the support count of C .**
- C. By ensuring that the size of itemset C does not exceed 2 items.
- D. By confirming that the Lift of itemset C is less than 1.0.

Question 5 (Creating):

The traditional Apriori algorithm suffers from extreme performance bottlenecks when mining large-scale logistics data due to generating millions of candidate itemsets. You are tasked with designing a modified framework that eliminates candidate generation entirely and only scans the database twice by compressing the transactional transaction logs into a frequent-pattern tree structure. Which algorithmic paradigm should you choose to build this framework?

- A. User-User Collaborative Filtering with Pearson Correlation.
- B. Ordinary Least Squares (OLS) Polynomial Regression transformation.
- C. FP-Growth (Frequent Pattern Growth) algorithm utilizing a prefix-tree structure.**
- D. Latent Dirichlet Allocation (LDA) with Gibbs Sampling.

Recommender System

Question 1 (Applying):

Project Eco-Routing deploys a food delivery feature on its platform. A user named Bob frequently orders high-calorie "Burgers" and "Pizzas" from vendors located within a 2-kilometer radius. If the platform utilizes a **Content-based Filtering** approach, which item is the system most likely to recommend to Bob next?

- A. A healthy organic salad from a newly registered restaurant 10 kilometers away that has zero ratings.
- B. A popular "Fried Chicken" dish (Categorized as Fast Food) from a local vendor located 1.5 kilometers away.**
- C. The most expensive seafood platter on the platform that is currently trending among high-income users.
- D. A random grocery item like dishwashing liquid to maximize the diversity of his feed.

Question 2 (Analyzing):

The platform expands its services to a brand-new residential district. A newly registered restaurant has just uploaded its menu, but no users have ordered from it or rated its items yet. Which specific challenge of Recommender Systems is occurring here, and what is the best strategy to mitigate it using the available data?

- A. Filter Bubble; mitigated by restricting users from seeing popular items.
- B. Data Sparsity; mitigated by applying Binary Cross-Entropy loss functions.
- C. New Item Cold Start Problem; mitigated by using Content-based Filtering to match the textual descriptions and categories of the new dishes with the historical preferences of users who like similar cuisines.**
- D. Overfitting; mitigated by increasing the number of hidden layers in a Matrix Factorization model.

Question 3 (Analyzing):

In User-based Collaborative Filtering, when predicting a rating for an active user on a target restaurant, the system calculates the similarity between users. Why is it a standard practice to subtract a user's average rating from their actual rating (creating a mean-centered rating) before calculating the weighted prediction score?

- A. To ensure that the final predicted rating always scales perfectly between 0 and 1.
- B. To compress the rating matrix and eliminate data sparsity challenges.
- C. To adjust for individual rating biases, since some users are naturally lenient (giving 5 stars easily) while others are strict (giving 2 stars for average service).**
- D. To force the system to prioritize Content-based attributes over historical rating profiles.

Question 4 (Evaluating):

During a sudden torrential downpour, the system must recommend optimized emergency shelter or rest points for the human couriers (shippers) to protect them from flooding. You have two recommendation engines available:

- Model X: High Precision (When it recommends a shelter, it is guaranteed to be safe and open, but it leaves out many nearby available shelters).
- Model Y: High Recall (It lists every single potentially available shelter in the area, but accidentally includes some locations that are already flooded or closed).

Given that couriers are actively driving in dangerous conditions with depleting phone batteries and cannot afford to travel to a closed location, which metric should you prioritize?

- A. Prioritize Model Y (High Recall) because it is always better to maximize the total number of options regardless of errors.
- B. Prioritize Model X (High Precision) because a False Positive (recommending a shelter that turns out to be closed) causes catastrophic real-world consequences for a courier stuck in a storm.**
- C. Prioritize an unpersonalized popularity model that lists the most reviewed restaurants in the city.
- D. Neither model is suitable; the system must randomly shuffle locations to maintain algorithmic fairness.

Question 5 (Creating):

To scale up the system to handle tens of millions of customers and items, the memory-based Collaborative Filtering approach becomes computationally impossible at run-time. Propose a model-based recommendation

architecture that maps both users and items into a shared, lower-dimensional latent factor space (e.g., of size K) to predict missing ratings via dot products, and specify the primary mathematical technique used.

A. Apriori algorithm utilizing a minimum confidence prune step.

B. Matrix Factorization (such as Singular Value Decomposition - SVD) optimized via Stochastic Gradient Descent (SGD) or Alternating Least Squares (ALS).

C. Fuzzy Inference System (FIS) utilizing Mamdani min-max composition.

D. Multiple Linear Regression with a Variance Inflation Factor (VIF) check.

Fuzzy Logic

Question 1

The Eco-Routing system utilizes human courier reports labeled "heavily congested" or "clear" (Data Source 2). Which Fuzzy Logic component is responsible for taking a precise, real-time traffic density sensor reading (e.g., 45 cars/km) and converting it into a degree of membership for the linguistic term "Heavily Congested"?

A. Defuzzifier

B. Inference Engine

C. Fuzzifier

D. Rule Base

Question 2

Two control rules contribute to the AEV's desired 'Minimum Speed' state: Rule 1 (Battery is Low) has a derived strength of 0.6, and Rule 2 (Motor Temp is High) has a derived strength of 0.9. If the output fuzzy sets are aggregated (combined) using the standard fuzzy Max operator, what is the height (degree of membership) of the resulting combined fuzzy set 'Minimum Speed'?

A. 0.6

B. 0.9

C. 1.5

D. 0.54

Question 3

The Project Eco-Routing team is choosing between a standard **Crisp** controller (fixed thresholds) and a **Fuzzy Logic** controller for the AEV's motor speed. If the control input (e.g., motor temperature) constantly crosses a sharp, fixed threshold, what is the major drawback of the Crisp Logic system compared to the Fuzzy system?

A. The crisp system will be significantly faster to compute.

B. The crisp system will cause abrupt and continuous oscillation (chattering) of the control action (e.g., emergency fan activation) near the boundary.

C. The Fuzzy Logic system will suffer from the combinatorial explosion of rules near the boundary.

D. Both systems will behave identically because temperature is a continuous variable.

Question 4

In Project Eco-Routing, the engineer uses a narrow (steep slope) Membership Function for the linguistic term "Battery is Low." What is the primary consequence of using a narrow Membership Function on the sensitivity and stability of the AEV's control system?

A. It makes the system robust against sensor noise, ignoring small changes.

B. It increases the necessary size of the rule base exponentially.

C. It increases the system's sensitivity, causing the AEV to switch control states rapidly and potentially leading to oscillatory behavior with small battery fluctuations.

D. It guarantees that the Defuzzifier will use the Center of Gravity method.

Question 5

Propose a Mamdani-type fuzzy rule for Project Eco-Routing designed to optimize the AEV's operational safety and resource management under worst-case scenarios, using the linguistic inputs: Traffic (High, Low) and Battery (Low, Adequate).

- A. IF Traffic IS High OR Battery IS Adequate THEN Route IS Aggressive.
- B. IF Traffic IS Low AND Battery IS Adequate THEN Route IS Standard.
- C. IF Traffic IS High AND Battery IS Low THEN Route IS Cautious (Detour/Recharge).**
- D. IF Traffic IS Low OR Battery IS Low THEN Route IS Aggressive.

Linear Regression

SECTION 2: Transfer / Retention Scenario

Linear Regression

CASE STUDY: SMART URBAN LOGISTICS AND GREEN DELIVERY FLEET (PROJECT ECO-ROUTING)

Question 1 (Applying): The Eco-Routing system estimates the Estimated Time of Arrival (ETA, in minutes, Y) based on Distance (km, X) using the Simple Linear Regression model: $Y = 3 + 1.8 \cdot X$. If an AEV has a distance of 15 km to cover, what is the predicted ETA?

- A. 27 minutes
- B. 32 minutes
- C. 30 minutes**
- D. 25.8 minutes

Question 2 (Analyzing): A Multiple Linear Regression (MLR) model predicts the AEV's Energy Consumption (Y, kWh) based on Motor Temperature (X1) and Average Speed (X2). If the coefficient beta 1 (or Motor Temperature) is 0.05, assuming all other factors remain constant, what is the statistical meaning of this finding?

- A. Motor Temperature has no relationship with Energy Consumption.
- B. A 1-unit increase in Motor Temperature increases the log-odds of Energy Consumption by 0.05.
- C. When the Motor Temperature increases by 1°C, the average Energy Consumption is expected to increase by 0.05 kWh.**
- D. 5% of the variation in Energy Consumption is explained by Motor Temperature.

Question 3 (Analyzing): Eco-Routing engineers observe that two input variables for the prediction model (Delay Time X1 and Traffic Level X2, both derived from the IoT Sensor Grid) are nearly perfectly correlated ($r > 0.95$). What severe statistical problem occurs when running Multiple Linear Regression (MLR) with these two variables?

- A. Autocorrelation in the time series.
- B. Multicollinearity, which causes the regression coefficients (β) to become unstable and difficult to interpret.**
- C. Heteroscedasticity, indicating that the error variance is non-constant.
- D. The slope of the model will always be zero.

Question 4 (Evaluating): An MLR model predicting customer satisfaction (Y, scale 1-5) has a very high Coefficient of Determination ($R^2 = 0.92$). However, upon inspecting the Residual Plot, the residual points form a distinct U-shape (parabolic pattern). Which evaluation is correct?

- A. The model is perfect because R^2 is high, and the U-shape is just random noise.
- B. The assumption of homoscedasticity is violated, requiring the use of Weighted Least Squares.
- C. The assumption of linearity is violated; the model is missing a non-linear relationship (e.g., quadratic) between the independent and dependent variables, requiring feature transformation.**
- D. The model is Underfitting (lack of complexity), requiring a reduction in the number of input variables.

Question 5 (Creating): Data from the IoT Sensor Grid shows that the relationship between Distance Traveled (X) and Remaining Battery Capacity (Y) is non-linear, where the rate of battery depletion slows down as distance increases (diminishing returns). To enable the use of standard Linear Regression, propose a common transformation for the independent variable X.

- A. Transform X to X^2 (Quadratic transformation).
- B. Transform Y to $\log Y$ (Logarithmic transformation for the dependent variable).
- C. Transform X to $\log X$ (Logarithmic transformation for the independent variable).**
- D. No transformation is needed because Linear Regression can handle non-linear relationships.

Logistic Regression

Question 1 (Applying): The Eco-Routing system predicts the probability of an AEV motor 'Overheating' ($Y=1$) based on real-time data from the IoT Sensor Grid. If the Logistic Regression model outputs a log-odds (logit) value of -0.8 for the current motor status, calculate the approximate probability (p) of the AEV motor overheating.

- A. 0.69
- B. 0.31**
- C. 0.50
- D. 0.45

Question 2 (Analyzing): A Logistic Regression model predicts the likelihood of a human courier receiving a 'Low Satisfaction Score' ($Y=1$) based on several factors, including 'Route Distance' (X). If the coefficient for Route Distance is $\beta = 0.25$ (positive), how should this coefficient be interpreted in terms of the risk?

- A. A 1 km increase in Route Distance increases the probability of a low score by 25%.
- B. A 1 km increase in Route Distance decreases the probability of a low score by 0.25.
- C. A 1 km increase in Route Distance increases the log-odds (logit) of receiving a low satisfaction score by 0.25, holding all other factors constant.**
- D. Route Distance is negatively correlated with customer satisfaction.

Question 3 (Understanding): The Eco-Routing system is required to predict the binary outcome 'Package Damaged' (Yes/No) based on AEV velocity and road conditions. Why is Logistic Regression, which uses the Sigmoid function, the appropriate choice over standard Linear Regression for this classification task?

- A. Linear Regression is too computationally intensive for continuous time-series data like velocity.
- B. The Sigmoid function bounds the output between [0, 1], allowing the result to be interpreted as a valid probability of damage, which Linear Regression's unbounded output cannot guarantee.**
- C. Logistic Regression is better suited for minimizing the Mean Squared Error (MSE).
- D. Logistic Regression automatically handles non-linear relationships without needing feature transformation.

Question 4 (Evaluating): The system uses Logistic Regression to predict 'Immediate AEV Failure' ($Y=1$) based on motor diagnostics. Given that the cost of a False Negative (failing to predict a failure that leads to a crash) is extremely high, which evaluation metric should the engineering team prioritize when setting the classification probability threshold?

- A. Precision (minimizing False Positives, or unnecessary maintenance checks).
- B. Recall (minimizing False Negatives, or missed actual failures).**
- C. Accuracy (overall correctness across all predictions).
- D. F1 Score (harmonic mean of Precision and Recall).

Question 5 (Creating): The logistics team wants to extend the predictive capability to classify delivery outcomes into three mutually exclusive levels: 'On Time,' 'Minor Delay,' and 'Major Delay.' What is the required architectural modification to adapt the framework from Binary Logistic Regression to handle this multi-class classification task?

- A. Replacing the Cross-Entropy loss function with Mean Squared Error (MSE).
- B. Using Ordinary Least Squares (OLS) minimization with an unbounded output.
- C. Employing Multinomial Logistic Regression (Softmax Regression) or training multiple One-vs-Rest (OvR) binary models.**
- D. Applying the Tanh activation function to normalize the input variables.

Latent Dirichlet Allocation (LDA)

Question 1 (Applying):

The data team applies LDA on the "Customer Feedback & Call Center Logs" (unstructured text data) to discover hidden topics in customer complaints. After training a 5-topic LDA model, one topic shows high probability words: *delay, rain, traffic, late, flooded*. Which of the following is the most accurate interpretation?

- A. Customers frequently mention these 5 words together in the same sentence.
- B. This topic represents a latent theme about "Weather-related Delivery Delays".**
- C. The model has high accuracy because it found 5 important keywords.
- D. This topic should be removed because it contains negative words.

Question 2 (Analyzing):

When running LDA on tens of thousands of customer reviews, the team notices that increasing the number of topics (K) from 5 to 20 causes the average coherence score to rise initially then drop sharply after $K=12$. What does this phenomenon indicate?

- A. The model is suffering from overfitting when K is too large.**
- B. The data only has exactly 12 real topics.
- C. LDA cannot handle large K values.
- D. The preprocessing step (stopword removal) was done incorrectly.

Question 3 (Analyzing):

The team compares two topics generated by LDA from courier call logs:

- Topic A: *battery, low, recharge, warning, stop* (high probability)
- Topic B: *friendly, fast, polite, helpful, smile* (high probability)

How should the team evaluate these two topics for operational improvement?

- A. Topic B is more important because it has more positive words.
- B. Topic A should be prioritized because it relates to AEV technical failures that affect service reliability.**
- C. Both topics are equally important since LDA treats all topics the same.
- D. Topic B is better because it has higher topic coherence.

Question 4 (Evaluating):

The engineering team wants to use LDA topic distributions as features for predicting customer satisfaction (1–5 stars). They notice that documents with high proportion in the “Damaged Package” topic consistently receive low satisfaction scores. Which statement is correct?

- A. This proves that LDA has better predictive power than Word Embedding.
- B. The “Damaged Package” topic distribution is a strong negative predictor for customer satisfaction.**
- C. They should only use the top 3 words of each topic instead of the full topic distribution.
- D. LDA cannot be used as features for downstream supervised learning.

Question 5 (Creating):

The current LDA model treats each customer review as a single document. However, delivery feedback contains multiple aspects (courier behavior, package condition, timing, etc.). Propose an improved modeling approach using LDA principles to better capture these multiple aspects.

- A. Use standard LDA with higher number of topics ($K=50$).
- B. Apply Bi-term LDA or Sentence-level LDA / Aspect-based LDA.**
- C. Switch to K-means clustering on TF-IDF vectors.
- D. Use only bigrams and trigrams in the vocabulary.

Deep Neural Networks (DNN)

Question 1 (Applying):

The team builds a Deep Neural Network to predict customer satisfaction score (1–5) using features from multiple data sources (order history, IoT telemetry, and text sentiment). This is a **regression** task. Which loss function is most appropriate?

- A. Binary Cross-Entropy
- B. Mean Squared Error (MSE)**
- C. Categorical Cross-Entropy
- D. Hinge Loss

Question 2 (Analyzing):

A DNN is trained to classify road conditions (Clear / Congested / Dangerous) from AEV dashcam images (Vision Stream). During training, the validation loss stops decreasing while training loss continues to drop significantly. What problem is occurring?

- A. Underfitting
- B. Overfitting**
- C. Vanishing Gradient
- D. Data Leakage

Question 3 (Analyzing):

The team uses a Deep Neural Network with 6 hidden layers to predict AEV energy consumption. They observe that adding more layers improves training accuracy but makes the model very slow and unstable during training. What is the most likely cause?

- A. The network is too shallow.
- B. Vanishing or Exploding Gradient Problem.**
- C. The learning rate is too high.
- D. They are using ReLU activation in all layers.

Question 4 (Evaluating):

The team has two models for predicting delivery delay (binary classification):

- Model X: Very deep DNN (many layers) – High training accuracy, moderate validation accuracy.
- Model Y: Shallower DNN with Residual Connections (ResNet-style) – Slightly lower training accuracy but much better validation accuracy and generalization.

Which model should be chosen for real-time deployment and why?

- A. Model X because deeper is always better.
- B. Model Y because residual connections help mitigate degradation problem and improve generalization.**
- C. Neither, they should use Logistic Regression instead.
- D. Model X because it has higher training accuracy.

Question 5 (Creating):

Design a suitable Deep Neural Network architecture for **multi-task learning** in Project Eco-Routing that simultaneously: (1) predicts delivery delay, (2) estimates customer satisfaction, and (3) detects unsafe road conditions from images.

- A. Three completely separate DNNs.
- B. A shared backbone network with multiple task-specific heads (Multi-Task Learning).**
- C. One very large DNN with 10 hidden layers for all tasks.
- D. Use only 1D-CNN because all data is time-series.

Word Embedding

Question 1 (Applying):

The team trains Word Embeddings on the customer feedback corpus. The vector for “delay” is very close to the vectors of “late”, “traffic”, and “rainy”. What does this closeness indicate?

- A. These words appear in the same sentences frequently and share similar semantic context.**
- B. These words have the same sentiment score.
- C. The model is biased toward negative words.
- D. These words have high TF-IDF scores.

Question 2 (Analyzing):

When using pre-trained Word Embeddings (e.g., Word2Vec or GloVe) trained on general English data versus embeddings trained specifically on the logistics customer review corpus, which one is generally better for understanding domain-specific terms like “satellite depot”, “last-mile”, “AEV”, and “gig-shipper”?

- A. General pre-trained embeddings are always better.
- B. Domain-specific embeddings trained on the company’s own corpus.**
- C. They will produce the same result.
- D. Character-level embeddings are better than word embeddings.

Question 3 (Analyzing):

The team observes that in the trained embedding space:

$\text{vector}(\text{"courier"}) - \text{vector}(\text{"human"}) + \text{vector}(\text{"AEV"})$ is close to $\text{vector}(\text{"autonomous"})$.

What capability of Word Embedding does this example demonstrate?

- A. Word Analogy / Semantic Reasoning**
- B. Sentiment Analysis
- C. Topic Modeling
- D. Dimensionality Reduction only

Question 4 (Evaluating):

The team wants to use Word Embeddings to automatically detect negative courier behavior from call transcripts. They average all word vectors in a review to get a document vector, then train a classifier. What is a major limitation of this approach?

- A. It completely ignores word order and context.**
- B. It always produces high accuracy.
- C. It cannot handle out-of-vocabulary words.
- D. It requires a very large embedding size (minimum 1000 dimensions).

Question 5 (Creating):

To better capture contextual meaning in customer complaints (e.g., the word “slow” can mean slow delivery or slow response depending on context), which advanced embedding technique should the team adopt instead of static Word2Vec/GloVe?

A. TF-IDF weighted Word2Vec

B. Contextual embeddings such as BERT or RoBERTa (contextualized word representations)

C. Increase the dimension of Word2Vec to 500

D. Use only FastText because it handles subwords

SECTION 3: BRIEF FOLLOW-UP SURVEY

Instructions: Please rate your level of agreement with the following statements on a scale from 1 to 5.

(1: Strongly Disagree | 2: Disagree | 3: Neutral | 4: Agree | 5: Strongly Agree)

Part 1: Continuous Learning & Knowledge Retention

1. In the past 2 months, I have actively read extra materials, research papers, or attended courses related to the 2 selected technical topics.
2. I still frequently apply or reference the foundational concepts of these 2 topics in my current projects, assignments, or daily technology discussions.

Part 2: Long-term AI Tool Usage

3. I have continued to use GenAI tools (e.g., ChatGPT, Gemini, Copilot) to explore and deepen my understanding of these technical topics over the last 2 months.
4. When encountering a new or difficult problem within these topics nowadays, my first instinct is still to consult an AI tool before looking elsewhere.

Part 3: Critical Reflection & Trust

5. After 2 months of independent learning, I feel more capable of analyzing and filtering out the biased or inaccurate technical responses generated by AI.
6. I now double-check AI-generated code or mathematical formulas against official documentation more rigorously than I did during the course.
7. My trust in AI's capacity to explain highly complex computer science models has consolidated and become more stable over time.

Part 4: Autonomy vs. Dependency

8. I feel that I have become less dependent on AI for straight answers, using it more as a brainstorming partner for technical logic.
9. Without AI tools, I would find it significantly harder to solve the structural and algorithmic problems of these 2 selected topics today.
10. The AI-supported learning habits I developed during the Intelligent Systems (IS) course have successfully transferred into my self-study routines for other university subjects.